SHEILLA MUMBI MACHARIA

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LinkedIn: My Profile | GitHub: My Profile | Portfolio

PROFESSIONAL SUMMARY

Data Science practitioner with a background in Applied Mathematics and hands-on experience in predictive modeling, analytics, and visualization. Skilled in Python, Tableau, and Streamlit for deriving actionable insights and building data-driven applications. Passionate about turning raw data into meaningful stories that drive smarter decisions, with proven success in projects spanning health prediction, retail analytics, and aviation risk assessment.

TECHNICAL SKILLS

Languages: Python, SQL, HTML/CSS
Libraries/Frameworks: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit
Tools: Excel, Tableau, Power BI, Jupyter Notebook, Git
Core Skills: Data Cleaning, Feature Engineering, Predictive Modelling, Data Visualization, Business Intelligence

PROJECTS

Pulse Metrix | Predictive Health Application | *Python, Streamlit, Scikit-learn, Pandas, HTML/CSS* | Repository

- Built a machine learning pipeline to predict 10-year coronary heart disease.
- Selected a Logistic Regression over a Gradient Boosting model achieving an accuracy of 70% vs 63%.
- Resolved class imbalance using SMOTE to prevent models from overfitting.
- Lowered the clinical threshold to 0.3 increasing Recall from 63% to 91% minimizing false negatives.

Kenya Regional Crop Yield Prediction | Machine Learning Pipeline | *Python, Scikit-learn, Pandas, NumPy, Matplotlib* | Repository

- Developed an agricultural forecasting pipeline to predict regional crop yields in Kenya by integrating climate data, pesticide usage, and historical production metrics
- Consolidated multi-source data from Kaggle, HarvestStat, and OpenAfrica, resolving spatial mismatches and engineering custom seasonal climate features (Long vs. Short Rains)
- Implemented and optimized Random Forest and XGBoost regressors, outperforming the baseline Linear Regression model by capturing complex, non-linear interactions between rainfall and crop output.
- Deployed the final model as a web application via Flask on Render, providing actionable forecasts to support early-warning food security planning and resource allocation.

Data Refinery Studio | Automated Data Preprocessing Web App | *Python, Streamlit, Pandas, NumPy, Plotly Express*| Repository

- Designed and built a self-service web workspace that automates data cleaning, structural diagnostics, and down-stream data preparation.
- Wrote robust backend scripts to prune duplicate records, trim string whitespaces, and automatically resolve mixed data types upon file upload.
- Programmed configurable workflows for numeric variables (Mean, Median, Zero) and categorical fields (Mode, 'Unknown') to handle structural missingness dynamically via user controls.
- Integrated real-time Plotly structural heatmaps to track before-and-after variance changes and engineered a memory-efficient cached utility block for instant CSV downloads.

SyriaTel Customer Churn Prediction | Predictive Analytics | *Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn* | Repository

- Built an end-to-end machine learning classification pipeline to predict customer attrition and safeguard telecom revenue using historical account data.
- Engineered data pipelines to drop non-actionable columns, encode binary plan metrics, and apply stratified splitting to manage class distributions cleanly across 3,333 records.
- Evaluated baseline models against Logistic Regression and a fine-tuned Decision Tree Classifier, choosing the tree architecture to achieve the best balance of interpretability and performance.

- Optimized the model to maximize Churn Recall (64%) and Precision (76%) at a 92% overall accuracy rate, successfully filtering high-risk accounts to help retention teams proactively minimize revenue loss.

Supermarket Performance Optimization Pipeline | Business Intelligence & Analytics | *Python (Pandas, NumPy, Statsmodels), SQL, Tableau, Power BI* | *Repository*

- Engineered an end-to-end retail data science framework to analyze the operational variables driving footprint efficiency and revenue velocity across store locations.
- Conducted bivariate OLS regression analysis to discover a deterministic linear relationship (ρ = 0.999$) between store area and inventory capacity, alongside identifying a traffic paradox where gross footfall decoupled from total sales volumes.
- Resolved traditional grouping errors by replacing rigid, hardcoded spatial boundaries with dynamic, dataset-driven percentiles, ensuring perfectly balanced classification metrics across varying store sizes.
- Built interactive dashboards in Tableau and Power BI to translate complex retail metrics into actionable stakeholder insights.

EDUCATION

Bachelor of Science in Applied Mathematics
University of Nairobi | Expected Graduation: 2026

Moringa School Data Science Bootcamp (August 2025 – February 2026)
Intensive hands-on program in Python, Machine Learning, and Data Visualization, culminating in real-world capstone projects (SyriaTel Customer Churn Prediction, Kenya Regional Crop Yield Prediction).

CERTIFICATIONS

Data Science Program – eMobilis Technological Institute
Comprehensive training in data science fundamentals, covering Python programming, data wrangling, exploratory data analysis, and introductory machine learning. Completed hands-on projects involving real-world datasets to develop skills in data cleaning, feature engineering, and insight generation for decision-making.

Python for Data Analysis – FreeCodeCamp
Focused training in Python for data analysis, including NumPy, pandas, and data visualization libraries. Applied techniques for data cleaning, transformation, and exploratory analysis to extract meaningful insights from structured datasets.

Machine Learning & Data Visualization – Kaggle
Hands-on learning in machine learning workflows and data visualization best practices using real-world datasets. Built predictive models, evaluated performance metrics, and created visual dashboards to communicate insights effectively.

LANGUAGES

English (Fluent) | Swahili (Fluent)